



A Colored Turan Proof of the Lollipop Formula

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June 9, 2026

Abstract

Let $a_{\mathcal{L}}(n)$ be the maximum number of connected regions into which n lollipops can divide the plane. We prove, from Paulsen's two geometric obstructions, the exact four-cluster formula

$$a_{\mathcal{L}}(n) = 4 \binom{n}{2} + S(n) + n + 1,$$

where

$$S(n) = \max_{a+b+c+d=n, 0 \leq a \leq b \leq c \leq d} \left(3 \sum_{1 \leq i < j \leq 4} m_i m_j - 2ab \right),$$

$$(m_1, m_2, m_3, m_4) = (a, b, c, d).$$

The proof is entirely combinatorial after the geometric reduction. It uses a colored Zykov symmetrization, a weighted blocker lemma, and an exact 3×4 matrix inequality. The matrix inequality is proved by support compression: alternating four-edge path compressions and double-star compressions reduce every integer matrix to star-forest normal form, where the efficient degree pattern is $K_{1,2} \cup K_2 \cup K_2$. This also explains the asymptotic cluster proportions

$$\frac{3}{16}, \frac{3}{16}, \frac{5}{16}, \frac{5}{16}.$$

1 The formula

A *lollipop* is a circle together with a half-line attached at one point of the circle and extending radially outward. We assume arrangements are generic: there are no tangencies, no triple intersection points, and no intersection occurs at an anchor point. This entails no loss for a maximum, since a non-generic arrangement can be perturbed without decreasing the number of regions.

Let C be the total number of crossings between distinct lollipops. Adding one generic lollipop with k crossings increases the number of regions by $k + 1$, and hence

$$\# \text{regions} = C + n + 1. \quad (1)$$

Thus the problem is to maximize C .

For $n \geq 0$, define

$$S(n) = \max_{a+b+c+d=n, 0 \leq a \leq b \leq c \leq d} \left(3 \sum_{1 \leq i < j \leq 4} m_i m_j - 2ab \right), \quad (m_1, m_2, m_3, m_4) = (a, b, c, d). \quad (2)$$

Equivalently, for a fixed multiset of four cluster sizes, the penalized pair is the pair with smallest product, i.e. the two smallest clusters.

Theorem 1. *For every $n \geq 0$,*

$$a_{\mathcal{L}}(n) = 4 \binom{n}{2} + S(n) + n + 1. \quad (3)$$

The first values obtained from (3) are

$$\begin{aligned} &1, 2, 10, 25, 45, 71, 104, 142, 186, 237, 294, 356, 425, 500, 580, 667, \\ &761, 859, 964, 1076, 1193, 1316, 1446, 1581, 1722, 1870, 2024, 2183, \\ &2349, 2521, 2698, 2882, 3073. \end{aligned}$$

2 Geometric input and the two-graph reduction

We use Paulsen's terminology. A pair of lollipops is *close* if the angle between the two stem directions is at most 90° . A pair is *intriguing* if the two circles either do not meet, or meet at angle at most 90° , after orienting the two circles in the same direction.

The following geometric facts are due to Cutler–Karlsson–Sloane and Paulsen. The present paper is a self-contained proof of the combinatorial extremal step that follows from these facts.

Lemma 2 (geometric lollipop input). *For a generic pair of lollipops:*

- (i) *any pair has at most 7 crossings;*
- (ii) *a close pair has at most 5 crossings;*
- (iii) *an intriguing pair has at most 5 crossings;*
- (iv) *a pair that is both close and intriguing has at most 4 crossings.*

Furthermore, among any four lollipops there is a close pair, and among any five lollipops there is an intriguing pair.

Remark 3. In the Lean formalization the geometric input is not treated as a single undifferentiated assumption. The strongest current upper boundary asks for distinct circle and ray-supporting-line facts for each unordered pair; from these facts Lean now constructs the finite

carrier-intersection witness and the automatic crossing table for the generic 7-crossing branch. The remaining upper inputs are named Euclidean route certificates which imply the component-wise 5- and 4-crossing savings for the close, intriguing, and close-and-intriguing branches. These routes include disjoint or nested circles, projection-distance or determinant/Cauchy separation of mixed circle-ray components, half-line outward separation when a ray starts outside a circle and points away from its center, and parallel-offset separation of ray-ray components. The close-direction interface is also split: Lean proves the trigonometric bridge from cyclic normalized closeness to a nonnegative dot product of bearing vectors, and records a normalized-bearing compatibility condition which turns this into a statement about the actual stored ray vectors. Since close radial pairs need not make one mixed circle-ray component empty, the Lean geometry folder also records a direct whole-carrier savings interface for coupled component-count proofs. Two direct coupled overlap theorems are now formalized: if two overlap witnesses together improve all four component estimates, then the whole carrier intersection has at most five points; if one point lies in all four components, then the whole carrier intersection has at most four points, under the standard noncoincident-circle and noncoincident-ray-line hypotheses. Lean also proves that a point lying in any three of the four lifted circle/ray product components automatically lies in the fourth component, so a triple-overlap witness gives the stronger four-point bound; likewise, a two-double-overlap certificate whose two witnesses coincide is converted to the same four-component bound. Lean uses the four-component form to turn the shared-anchor close-route audit example into an actual 4-crossing direct carrier-savings certificate. Lean converts the routes to component-savings bounds and assembles the automatic pair certificates into the upper input used by the colored-Turan proof. It also has a direct-savings automatic boundary whose close and intriguing inputs are whole-carrier cardinality certificates, so the coupled overlap theorems can feed the final Theorem 1 boundary without first being decomposed into four independent component caps. The theorem-facing subtheorem index now also exposes the route transport lemmas explicitly: named 5- and 4-crossing routes convert to component-savings certificates, direct carrier-savings certificates, and the corresponding local or global rational pair-crossing bounds. The separate Lean modules `CompleteFormalization/FiniteCarrier.lean` and `CompleteFormalization/AutomaticUpper.lean` contain this reduction, and `CompleteFormalization/Proof.lean` indexes the resulting theorem boundary and proves Theorem 1 from it. A separate `EndToEndFormalization/` folder also exports theorem endpoints whose lower input is finite carrier-subset data for the produced blow-ups, so the upper and lower automatic-carrier reductions can be combined without changing the older proof stack; its `Statement.lean` and `Subtheorems.lean` files record the statement, the named proved bridge lemmas, and the most concrete current certificate boundary in one place. In that boundary, triple-overlap, two-double-overlap, and all-four-overlap certificates are converted by Lean into the direct whole-carrier savings needed for the close/intriguing branches, with triple-overlap exposed as a direct 4-crossing savings route and coincident two-double overlap exposed the same way. The lower side can also be stated using cardinality lower bounds on the automatic carrier finsets themselves; when such a bound is supplied, Lean uses the automatic finset as the local lower witness. The current theorem-facing statement uses Lean’s canonical sorted-quad cluster witness internally, so this boundary no longer exposes a separate cluster-map witness as an input. The all-four-overlap interface also has a common-anchor constructor: if two primitive lollipops share their anchor point, Lean proves that point lies in all four lifted circle/ray components and hence supplies the direct 4-crossing savings certificate under the usual noncoincidence hypotheses. The same interface now has a

primitive-point constructor: any coordinate point certified to lie on both primitive circles and both primitive rays produces the all-four and five-overlap witnesses, and hence the corresponding direct 4- and 5-crossing savings certificates under the same noncoincidence hypotheses. It also has a primitive triple-overlap constructor: ordinary coordinate membership facts showing that one point lies in any three of the four circle/ray product components are converted to all-four overlap, primitive five-overlap, and direct 4- and 5-crossing savings certificates. The two-double-overlap interface now has primitive coordinate constructors as well: Lean accepts ordinary circle/ray membership facts for two coordinate points in each of the three component-pair patterns, builds the lifted two-double witness, and derives the direct 5-crossing savings certificate. When the two-double witnesses are the same primitive coordinate point, Lean also derives the all-four overlap witness and the direct 4-crossing savings certificate. These 4-crossing primitive witnesses are bundled in a flexible primitive four-overlap boundary, so the close-and-intriguing branch may use all-four, triple, or same-point two-double coordinate data before converting to the existing primitive upper certificate. Lean exposes this flexible primitive upper boundary with the canonical automatic-cardinality lower boundary, and with the component-indexed, table-exact, finset-exact, component-covered, component-finset, disjoint component-finset, and indexed disjoint component-finset lower endpoints. The clean statement/proof index also re-exports the exact indexed-carrier, finite-carrier, common-anchor lower, automatic-cardinality, and canonical Karlsson lower-table bridge lemmas, re-exports the route/direct-savings/overlap upper transport lemmas, and exposes the common-anchor and raw primitive all-four/five direct-savings aliases at the theorem-facing layer. Its split boundary now also has overlap, primitive, and flexible primitive component-indexed and indexed-disjoint-component endpoints, where the lower construction supplies disjoint component point families, with or without exact component-carrier coverage. At the stepwise family level, Lean also has a primitive overlap upper certificate whose close/intriguing fields are these raw coordinate overlap witnesses rather than prebuilt lifted component-overlap objects. This primitive upper certificate converts to the existing automatic overlap boundary, and the theorem-facing endpoint combines it with exact indexed carrier enumerations plus one equality to the canonical Karlsson lower table. Direct carrier-savings certificates are also monotone in their bound in Lean, so a stronger certificate, such as a 4-crossing saving, can be reused wherever the upper proof only asks for a 5-crossing saving. The Euclidean derivation of the route certificates for a concrete lollipop model remains part of the model-specific geometric work.

Given an arrangement of n lollipops, define graphs D, E on the same vertex set by

$$\begin{aligned} ij \in D &\iff L_i, L_j \text{ are not close,} \\ ij \in E &\iff L_i, L_j \text{ are not intriguing.} \end{aligned}$$

By Lemma 2,

$$D \text{ is } K_4\text{-free,} \quad E \text{ is } K_5\text{-free.} \quad (4)$$

For a pair ij , Lemma 2 gives the pointwise estimate

$$c_{ij} \leq 4 + \mathbf{1}_{ij \in D} + \mathbf{1}_{ij \in E} + \mathbf{1}_{ij \in D \cap E}. \quad (5)$$

Therefore

$$C \leq 4 \binom{n}{2} + \sigma(D, E), \quad \sigma(D, E) := |D| + |E| + |D \cap E|. \quad (6)$$

Theorem 1 is now reduced to the following colored Turan theorem.

Theorem 4 (colored Turan theorem). *If D is K_4 -free and E is K_5 -free on the same n -vertex set, then*

$$\sigma(D, E) \leq S(n). \quad (7)$$

The proof occupies Sections 3–5. The lower construction is given in Section 6; together with (1) and (6), it proves Theorem 1.

3 Colored Zykov symmetrization

Color every unordered pair of vertices with one of four colors:

$$0 : \text{neither } D \text{ nor } E, \quad A : E\text{-only},$$

$$B : D\text{-only}, \quad X : D \cap E.$$

These colors have weights

$$w(0) = 0, \quad w(A) = 1, \quad w(B) = 1, \quad w(X) = 3,$$

so that $\sigma(D, E)$ is the total color-weight.

Lemma 5 (colored Zykov reduction). *An extremal pair (D, E) for σ may be chosen as a blow-up of a complete colored quotient: the vertex set is partitioned into classes $V_1, \dots, V_k$, every pair inside one class has color 0, and every pair between two distinct classes has one fixed color from $\{A, B, X\}$. In particular, there are no color-0 pairs between distinct classes.*

Proof. Suppose uv has color 0. Let $d_w(v)$ be the total color-weight from v to all vertices other than u, v . If $d_w(u) \leq d_w(v)$, replace u by a clone of v : for every $z \neq u, v$, give uz the same color as vz , while keeping uv color 0. This operation changes σ by $d_w(v) - d_w(u) \geq 0$.

It cannot create a K_4 in D . Indeed, a new D -clique using the clone u and not v corresponds to the same clique with v in place of u , while a clique using both u and v is impossible because uv is not a D -edge. The same argument forbids creation of a K_5 in E . Thus the operation preserves (4) and does not decrease σ .

More explicitly, among all pairs satisfying (4) and maximizing σ , choose one that maximizes the ordered twin-pair potential

$$\Phi = \sum_v |\{z : \text{the color-row of } z \text{ equals the color-row of } v\}|.$$

For a color-0 pair uv , maximality of σ forces $d_w(u) = d_w(v)$, since cloning in either direction is allowed. Hence either clone keeps σ maximal. If u and v are not already twins, clone the vertex with the smaller twin class to the one with the larger twin class. This strictly increases Φ : the cloned vertex joins the larger twin class, while every other twin class loses at most the cloned vertex. This contradicts the choice of the tie-breaker. Therefore every color-0 pair is a twin pair. Consequently color-0 is an equivalence relation, its classes are twin classes, and

between two distinct classes all pairs have one fixed nonzero color. This is the claimed quotient form. $\square$

Let the quotient class sizes be $x_1, \dots, x_k$, with $\sum_i x_i = n$, and put

$$Q = \sum_i x_i^2.$$

Let

$$a = \sum_{ij \in A} x_i x_j, \quad b = \sum_{ij \in B} x_i x_j,$$

where A and B now also denote the graphs of A -colored and B -colored quotient edges. The total cross-class mass is

$$P = \sum_{i < j} x_i x_j = \frac{n^2 - Q}{2}.$$

Since X -edges have weight 3, while A - and B -edges have weight 1,

$$\sigma = 3P - 2a - 2b = \frac{3}{2}(n^2 - Q) - 2a - 2b. \quad (8)$$

The forbidden cliques translate into blocker conditions. Since D consists of colors $B \cup X$, every four quotient vertices contain an A -edge; otherwise they form a K_4 in D . Thus

$$\alpha(A) \leq 3. \quad (9)$$

Similarly, since E consists of colors $A \cup X$,

$$\alpha(B) \leq 4. \quad (10)$$

4 The weighted blocker lemma

For nonnegative integer weights $x_1, \dots, x_k$ with total n , define $Q = \sum_i x_i^2$. If G is a graph on $[k]$, write

$$w(G) = \sum_{ij \in E(G)} x_i x_j.$$

For $r \geq 1$, let

$$\rho_r(x) = \min_{\mathcal{P}} \sum_{P \in \mathcal{P}} \left(\sum_{i \in P} x_i \right)^2,$$

where the minimum is over all partitions of $[k]$ into r possibly empty parts.

Lemma 6 (weighted Turan theorem). *If G is K_{r+1} -free, then*

$$w(G) \leq \frac{1}{2}(n^2 - \rho_r(x)). \quad (11)$$

Proof. The usual Zykov symmetrization works with vertex weights: if two nonadjacent vertices have different weighted neighborhoods, clone the smaller weighted neighborhood to the larger

one. This preserves K_{r+1} -freeness and does not decrease $w(G)$. In an extremal graph, two nonadjacent vertices of positive weight must therefore have the same weighted neighborhood. It follows that nonadjacency is transitive on the positive-weight vertices: otherwise two nonadjacent positive vertices would have different adjacency to a third positive vertex. Hence the induced graph on the positive-weight support is complete multipartite. Since G is K_{r+1} -free, this positive-weight multipartite graph has at most r parts; vertices of weight zero may be assigned to the parts arbitrarily. For a partition with part weights $s_1, \dots, s_r$, its edge weight is

$$\sum_{i < j} s_i s_j = \frac{1}{2} \left(n^2 - \sum_i s_i^2 \right),$$

which is maximized exactly when $\sum_i s_i^2$ is minimized. $\square$

Define

$$M(n) = \min_{a+b+c+d=n, a,b,c,d \in \mathbb{Z}_{\geq 0}} \left[\frac{3}{2}(a^2 + b^2 + c^2 + d^2) + 2ab \right]. \quad (12)$$

Since

$$\frac{3}{2}n^2 - \left[\frac{3}{2}(a^2 + b^2 + c^2 + d^2) + 2ab \right] = 3 \sum_{i < j} m_i m_j - 2ab,$$

where $(m_1, m_2, m_3, m_4) = (a, b, c, d)$, and since the two variables in the penalized product may be labelled as the two smallest cluster sizes, the definitions imply

$$S(n) = \frac{3}{2}n^2 - M(n). \quad (13)$$

Lemma 7 (weighted blocker lemma). *Let A, B be graphs on $[k]$ with $\alpha(A) \leq 3$ and $\alpha(B) \leq 4$. With the notation above,*

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq M(n). \quad (14)$$

Proof. The complement of A is K_4 -free. By Lemma 6,

$$\frac{1}{2}(n^2 - Q) - w(A) \leq \frac{1}{2}(n^2 - \rho_3(x)),$$

and hence

$$w(A) \geq \frac{1}{2}(\rho_3(x) - Q). \quad (15)$$

Likewise, the complement of B is K_5 -free, so

$$w(B) \geq \frac{1}{2}(\rho_4(x) - Q). \quad (16)$$

Combining (15) and (16),

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq \rho_3(x) + \rho_4(x) - \frac{1}{2}Q. \quad (17)$$

Choose partitions attaining $\rho_3(x)$ and $\rho_4(x)$, and intersect them. This gives a 3×4 nonnegative integer matrix $U = (u_{ij})$, where u_{ij} is the total weight lying in the i -th part of the 3-partition

and the j -th part of the 4-partition. Let

$$r_i = \sum_j u_{ij}, \quad c_j = \sum_i u_{ij}.$$

Then

$$\rho_3(x) = \sum_i r_i^2, \quad \rho_4(x) = \sum_j c_j^2.$$

Also

$$Q = \sum_\ell x_\ell^2 \leq \sum_{i,j} u_{ij}^2,$$

because squaring after grouping can only increase the sum of squares. Thus (17) gives

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq \sum_i r_i^2 + \sum_j c_j^2 - \frac{1}{2} \sum_{i,j} u_{ij}^2.$$

The right side is at least $M(n)$ by the matrix theorem proved in Section 5. $\square$

We now prove Theorem 4, assuming the matrix theorem for a moment.

Proof of Theorem 4. By Lemma 5, assume (D, E) is in quotient form. Then (9) and (10) allow Lemma 7, so

$$\frac{3}{2}Q + 2a + 2b \geq M(n).$$

Using (8),

$$\sigma = \frac{3}{2}n^2 - \left(\frac{3}{2}Q + 2a + 2b\right) \leq \frac{3}{2}n^2 - M(n) = S(n),$$

by (13). This proves (7). $\square$

5 The 3×4 matrix theorem

For a 3×4 nonnegative integer matrix $U = (u_{ij})$ with total sum n , row sums r_i , and column sums c_j , put

$$F(U) = \sum_{i=1}^3 r_i^2 + \sum_{j=1}^4 c_j^2 - \frac{1}{2} \sum_{i,j} u_{ij}^2. \quad (18)$$

Theorem 8 (matrix theorem). *For every 3×4 nonnegative integer matrix U of total sum n ,*

$$F(U) \geq M(n). \quad (19)$$

The proof is the main technical point. Think of the positive entries of U as edge masses on the bipartite graph $K_{3,4}$, whose left vertices are the rows and whose right vertices are the columns. If two positive cells share a row or a column, call the corresponding edges adjacent. Expanding (18) gives

$$F(U) = \frac{3}{2} \sum_e u_e^2 + 2 \sum_{e \sim f} u_e u_f. \quad (20)$$

Here the second sum is over unordered pairs of distinct adjacent support edges.

Compression to a star forest

The next lemma is the normal-form step needed below: every matrix can be compressed to a star forest without increasing F . Equality may occur during compression, so the conclusion concerns the compressed matrix rather than the original support.

Lemma 9 (support compression). *For every nonnegative integer matrix U , there is a nonnegative integer matrix U' with the same total sum such that*

$$F(U') \leq F(U)$$

and the support graph of U' is a forest whose connected components are stars.

Proof. We repeatedly apply a support-decreasing move. Since the support has at most twelve edges, it is enough to show that every non-star support admits one such move preserving the total mass and not increasing F . In each alternating move below, ε is restricted to the integer interval on which all changed entries remain nonnegative. This interval contains 0, and its two endpoints are obtained by killing at least one entry on one of the two alternating classes. Hence endpoint moves keep the matrix integral, do not create new support edges, and strictly decrease the support unless the matrix was already outside the case under discussion.

The graph-theoretic bookkeeping used here is finite. In $K_{3,4}$, every non-star support, after relabeling rows and columns, falls into one of the following descent shapes: a four-cycle, a four-edge path in one of the two orientations, or a diameter-3 double-star with its remaining leaves in the corresponding double-star envelope. The Lean formalization proves this classification by exhaustive checking over the twelve possible support edges; the paragraphs below prove the descent move for each canonical shape.

First suppose the support contains a cycle. Since the support graph is bipartite, the cycle is even. Add ε and $-\varepsilon$ alternately around the cycle. Row sums and column sums, and therefore the first two terms of (18), remain fixed. The final term $-\frac{1}{2} \sum u_{ij}^2$ is a concave quadratic in ε . The allowed integer interval is obtained by requiring the entries on the decreasing half of the cycle to stay nonnegative in each direction. One endpoint of this interval does not increase F : if both endpoints had larger value than at 0, concavity would force the value at 0 to be larger than itself by interpolation between the endpoints. The chosen endpoint makes at least one formerly positive entry zero and creates no new positive entries, because every edge being changed already lies in the support.

Thus we may next assume that the support is a forest. If a tree component has diameter at least 4, choose four consecutive edges on a path in that component, with masses x_1, x_2, x_3, x_4 . Replace them by

$$x_1 + \varepsilon, \quad x_2 - \varepsilon, \quad x_3 + \varepsilon, \quad x_4 - \varepsilon.$$

The total mass is preserved. The three internal vertex sums on this four-edge path are unchanged; only the two endpoint vertex sums change. In the expression (20), the coefficient of ε^2 is

$$2 - \frac{1}{2} \cdot 4 = 0,$$

so F is linear in ε on the allowed integer interval. One endpoint of that interval does not increase F according to the sign of this linear function, and kills at least one of the four

entries, again without creating a new support edge.

It remains to treat a forest component of diameter exactly 3, i.e. a double-star. This is the only remaining non-star case: a forest with no path of four edges and no component of diameter 3 has every component of diameter at most 2, hence is a star forest. Let y be the mass of the central edge. Choose a leaf edge of mass x on one side and a leaf edge of mass z on the other. Let A be the sum of the other leaf-edge masses incident with the center on the x -side, and let B be the analogous sum on the z -side. If we delete the central edge and add its mass y to the x -edge, a direct calculation from (20) gives the change

$$F_{\text{old}} - F_x = y(2B + 2z - x).$$

If instead we add y to the z -edge, then

$$F_{\text{old}} - F_z = y(2A + 2x - z).$$

At least one of these two quantities is nonnegative: otherwise $x > 2B + 2z \geq 2z$ and $z > 2A + 2x \geq 2x$, impossible. Hence one of the two operations preserves the total mass, keeps all masses integral and nonnegative, deletes the central edge, and does not increase F . Repeating this step removes all diameter-3 components.

Each move strictly decreases the number of positive entries, and the support cardinality is a nonnegative integer. Therefore the process terminates, and when it terminates no cycle, no four-edge path in a tree, and no double-star component remains. A forest component with no path of four edges and no diameter-3 double-star has diameter at most 2, so it is a star. The final support is therefore a star forest, proving the lemma. $\square$

Star forests

For a star component with t positive edges of masses $z_1, \dots, z_t$ and total mass $s = \sum_{\ell} z_{\ell}$, its contribution to F is

$$s^2 + \frac{1}{2} \sum_{\ell=1}^t z_{\ell}^2. \quad (21)$$

In particular,

$$s^2 + \frac{1}{2} \sum_{\ell=1}^t z_{\ell}^2 \geq \left(1 + \frac{1}{2t}\right) s^2 = \frac{s^2}{\eta_t}, \quad \eta_t := \frac{2t}{2t+1}. \quad (22)$$

Lemma 10 (star-forest minimum). *If U is a nonnegative integer matrix whose support is a star forest, then $F(U) \geq M(n)$, where n is the total sum of its entries.*

Proof. If $n = 0$, then $F(U) = M(0) = 0$. Hence assume $n > 0$. Let the star components have degrees t_{ν} and total masses s_{ν} . If the degree list is not $(2, 1, 1)$, then

$$\Gamma := \sum_{\nu} \eta_{t_{\nu}} \leq 2. \quad (23)$$

Indeed, one or two components give $\Gamma < 2$, since each $\eta_t < 1$. If there are three components, then the seven vertices of $K_{3,4}$ allow only $(2, 1, 1)$ or $(1, 1, 1)$; excluding $(2, 1, 1)$ leaves $(1, 1, 1)$, for which $\Gamma = 3 \cdot (2/3) = 2$. Here a star with t positive edges uses $t + 1$ support vertices. Thus

three nonempty components with largest degree at least 3, or with two components of degree at least 2, would require at least eight support vertices, impossible in $K_{3,4}$. There cannot be four nonempty star components, because each such component contains at least one row vertex and the components are disjoint while there are only three rows.

By (22) and Cauchy's inequality,

$$F(U) \geq \sum_{\nu} \frac{s_{\nu}^2}{\eta_{t_{\nu}}} \geq \frac{(\sum_{\nu} s_{\nu})^2}{\sum_{\nu} \eta_{t_{\nu}}} = \frac{n^2}{\Gamma} \geq \frac{n^2}{2}.$$

For $n \geq 4$, the balanced quadruple in (12) gives $M(n) \leq n^2/2$. Explicitly, for $n = 4q + r$, use (q, q, q, q) , $(q, q, q, q+1)$, $(q, q, q+1, q+1)$, or $(q, q+1, q+1, q+1)$ according as $r = 0, 1, 2, 3$. The resulting value is at most $n^2/2$ for $n \geq 4$. For $n \leq 3$, the identity

$$F(U) = \frac{3}{2}n + 3 \sum_{i,j} \binom{u_{ij}}{2} + 2 \sum_{\substack{(i,j) < (i',j') \\ (i=i' \text{ or } j=j'), (i,j) \neq (i',j')}} u_{ij} u_{i'j'}$$

shows $F(U) \geq \frac{3}{2}n = M(n)$. Thus every non- $(2, 1, 1)$ star forest satisfies the desired bound.

Now suppose the degree list is $(2, 1, 1)$. The degree-two star must be centered at a row vertex: if it were centered at a column vertex, only one row vertex would remain for the two disjoint K_2 components. Thus, up to row and column permutation, its four positive edge masses are

$$\begin{pmatrix} a & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & d \end{pmatrix}.$$

For this matrix,

$$F(U) = (a+b)^2 + c^2 + d^2 + \frac{1}{2}(a^2 + b^2 + c^2 + d^2) = \frac{3}{2}(a^2 + b^2 + c^2 + d^2) + 2ab.$$

Since $a + b + c + d = n$, this is at least $M(n)$ by definition. $\square$

Proof of Theorem 8. Apply Lemma 9 to replace U by a star-forest matrix U' with the same total sum and $F(U') \leq F(U)$. Lemma 10 gives $F(U') \geq M(n)$. Hence $F(U) \geq M(n)$. $\square$

The Lean formalization follows this proof split at the level of checked subtheorems: the proved support-descent theorem and the proved star-forest minimum theorem give the proved matrix theorem. The support-descent proof uses the finite relabeling/exhaustion described above, and the star-forest proof is checked by exhausting the canonical star-forest support cases. More precisely, Lean proves the finite star-forest shape classification, the non-star descent classification, the one-step descent theorem, the canonical case package, the star-forest minimum, and the final matrix theorem under the names

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isSupportStarForest_iff_canonicalShape,
nonstar_support_has_descent_shape,
support_descent_step_proven,
canonicalStarForestMinimumCases_proven,
starForestMinimum_proven,
matrix_theorem_proven.

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The separate theorem-facing assembly files import these proved Section 5 endpoints into the displayed Theorem 1 statement. Thus the matrix theorem is not a remaining Lean input to Theorem 4.

6 The lower construction

The lower construction uses Karlsson’s optimal four-lollipop configuration from [1]. We use only the following crossing table for that base configuration. Label the four base lollipops so that their pairwise crossing table is

	1	2	3	4
1	*	5	7	7
2		*	7	7
3			*	7
4				*

Thus the pair (1, 2) contributes 5 crossings and the other five pairs contribute 7 crossings.

Lemma 11 (blow-up stability). *Assume the four-lollipop base configuration with the crossing table displayed above. For any nonnegative integers m_1, m_2, m_3, m_4 , there is a generic arrangement of $m_1 + m_2 + m_3 + m_4$ lollipops whose crossing number is*

$$L(m_1, m_2, m_3, m_4) = 4 \sum_i \binom{m_i}{2} + 5m_1m_2 + 7 \sum_{\substack{i < j \\ (i,j) \neq (1,2)}} m_i m_j. \quad (24)$$

Proof. Replace the i -th base lollipop by m_i sufficiently small, mutually generic perturbations. Transverse crossings between two distinct base lollipops persist under sufficiently small perturbation in the bounded circle–circle and mixed circle–ray components. For the unbounded ray components this is not a compactness consequence by itself: small changes of ray directions can create intersections far from the base configuration. The perturbation must therefore be chosen with explicit coordinate inequalities certifying the full primitive carrier intersection for every affected pair (equivalently, with the local finite-carrier certificates used in the Lean boundary). Under those local certificates, every pair from two different clusters has the same crossing number as the corresponding base pair.

Inside one cluster, choose the copies in a tiny one-parameter family obtained by small translations and small independent rotations of the stem direction. For two sufficiently close copies of one lollipop, the two circles meet in two points, each stem meets the other circle once, and the two stems do not meet; these four intersections are transverse and persist under a generic small perturbation. Therefore every intra-cluster pair contributes exactly 4 crossings. A final

arbitrarily small perturbation removes accidental triple points without changing any pairwise crossing count. $\square$

Remark 12. In the Lean formalization this lower construction is represented by the theorem-facing certificate package

`KarlssonBaseBlowUpIncrementalLowerData`

and, more locally, by

`KarlssonBaseSixPairBlowUpIncrementalLowerData,`

where the base table is supplied as the six displayed unordered entries plus symmetry. The checked part expands that six-entry table to the universal base-pair table and can also assemble the copy-pair crossing values from one local certificate for each blown-up pair. It then converts the construction to the pairwise lower interface, sums the inherited cluster table, and derives the ordered-insertion region equation. The current Lean boundary also has a stepwise form in which each insertion step is supplied locally and Lean assembles the recurrence; for the exact OEIS four-base table, the displayed $1 \rightarrow 2 \rightarrow 8 \rightarrow 23 \rightarrow 45$ region arithmetic is recorded in this stepwise form. The complete Lean index also exports the six-entry base-table certificate, the unordered base sum 40, the resulting base region count 45, and the canonical four-fiber cluster witness used in the finite pair-counting collapse. For the exceptional exact OEIS pair (Q_0, Q_1) , Lean also checks the cyclic-close fact, Paulsen’s strict obtuse-intersection condition showing the pair is not intriguing, and a half-line-outward route proving the local ≤ 5 crossing and finite-witness-cardinality bounds from the six-pair base-coordinate certificate. Lean also proves that all four exact OEIS base qoppas have normalized-bearing compatibility, so the recorded normalized directions agree with the actual coordinate ray vectors. The separate complete-formalization OEIS geometry file now also names an explicit ray-ray intersection point and proves that the mixed component $\text{ray}(Q_0) \cap \text{circle}(Q_1)$ consists only of the two x-axis points with coordinates $(9 \pm \sqrt{57})/20$, forcing any exact six-pair finite witness to contain at least those two points. It also solves the circle-circle component: the exact radical-axis relation $20045x + 40y = 3$ and the two radical-coordinate circle-circle witnesses are proved in Lean, and together with the ray-ray and ray-circle points they give a five-point finset whose coercion is exactly the primitive (Q_0, Q_1) carrier intersection. Thus Lean now constructs the local (Q_0, Q_1) coordinate crossing certificate directly. For every non-exceptional exact base pair (Q_0, Q_2) , (Q_0, Q_3) , (Q_1, Q_2) , (Q_1, Q_3) , and (Q_2, Q_3) , the complete Lean folder also constructs all seven component points, proves their carrier memberships and distinctness, derives automatic finite-witness cardinality 7, and exports the local coordinate crossing certificates. The end-to-end Lean folder now packages these six exact automatic carrier-cardinality facts as the Nat-valued OEIS base lower table: the exceptional base pair contributes 5, and the five non-exceptional base pairs contribute 7. Lean also relabels the exact OEIS base arrangement by the canonical all-ones cluster witness and proves the same automatic lower table in those canonical labels. In this relabeled base case, Lean also uses the automatic carrier finset itself as the local lower subset and constructs the local monotone lower certificate consumed by the pairwise lower interface. The Euclidean realization of the local blow-ups with exactly these pairwise crossing values remains a model-specific geometric certificate, not a hidden algebraic assumption. In particular, the unbounded ray cases in Lemma 11 require concrete coordinate

or local perturbation certificates; they do not follow from a bare compactness argument. Lean now also includes a monotone pairwise lower interface:

PairwiseCardinalityClusteredKarlssonBlowUpIncrementalLowerBoundData,

where a concrete construction only has to prove that each blown-up pair has at least the corresponding Karlsson table value. Lean sums these pairwise inequalities over unordered pairs and derives a lower-bound attainment statement from the ordered region recurrence. The local geometric bridge is also formalized in **PrimitiveGeometry/LowerWitness.lean**: a finite set of distinct points inside one carrier intersection, together with the ordinary local carrier-crossing certificate for the full pair intersection, gives the required local monotone copy-pair lower certificate. The new **IndexedLowerWitness** file turns an injective indexed family of k explicit carrier points into a k -point lower subset and hence, with a local carrier certificate, into a $\text{pairCross} \geq k$ lower bound. The companion **IndexedCarrier** file handles the exact-enumeration form: if an injective indexed family has image equal to the whole primitive carrier intersection, Lean converts it into the local or pairwise finite-carrier crossing certificate, computes the corresponding automatic finite-carrier cardinality and automatic crossing-table entry, and also recovers the corresponding lower subset. Lean also proves that this exact image equality can be supplied by showing indexed-point membership plus coverage of all carrier points, or by covering the four primitive component cases circle-circle, circle-ray, ray-circle, and ray-ray separately. The **AutomaticCardinalityWitness** file pushes the same indexed point data directly to the automatic carrier finset cardinality inequality used by the strongest lower boundary. The theorem-facing statement also exposes an **IndexedPointCertificate** endpoint: with overlap-witness upper data, explicit indexed lower point families of the canonical 4, 5, and 7 sizes imply Theorem 1. Its **ComponentIndexedPointCertificate** variant is a lower-bound-only component version: four disjoint component point families are enumerated as one canonical lower-point family, without requiring the construction to prove that those points cover the whole carrier intersection. Its **ExactIndexedCarrierCertificate** variant strengthens the lower input to exact indexed enumerations of the whole canonical primitive pair carriers and derives the indexed-point boundary internally. The **TableExactIndexedCarrierCertificate** variant reduces the remaining pair-table bookkeeping to one equality with the canonical rational Karlsson lower table, from which Lean derives the increasing-entry and non-increasing-entry facts. A finset-exact carrier variant lets the construction supply each exact finite carrier set directly, together with its set equality and cardinality, instead of manufacturing an indexed enumeration. Lean also proves that this finite-set equality can be derived from finite-set membership plus carrier coverage, or from separate coverage of the four circle/ray component cases; these helpers identify the supplied finite set with the automatic carrier finset and compute the corresponding automatic cardinality and crossing-table entry. The theorem-facing Lean layer now also exposes this component-covered finite-carrier form as a full endpoint: the construction supplies the finite set, membership, the four component-coverage fields, and the cardinality, while Lean derives exactness internally. A still more component-local endpoint lets the construction supply four finite sets, one for each of the circle-circle, circle-ray, ray-circle, and ray-ray components; Lean unions those sets and derives the same component-covered carrier boundary. A disjoint component-finet endpoint now also derives that union cardinality from pairwise disjointness, four component cardinalities, and a single component-size sum. An indexed disjoint component-finet endpoint goes one step closer to coordinate verification by taking each component set to be the image of an injective indexed point fam-

ily, but this exact endpoint still includes component coverage of the whole carrier. The new **LowerAnchorWitness** file proves the basic shared-anchor case of this bridge: a common anchor is an actual primitive carrier-intersection point, giving a one-point lower subset and, with a local carrier certificate, a **pairCross** ≥ 1 bound. This does not by itself supply the required 4, 5, or 7 lower multiplicities. The separate **AutomaticLower** module composes this with the automatic finite-carrier table: if the produced pair table agrees with the automatic table and the construction supplies these lower subsets, Lean assembles the monotone lower certificate used by Theorem 1. It also contains a specialized Nat-valued form of the Karlsson lower table, where the construction supplies subsets of sizes 4, 5, or 7 directly and Lean proves that this natural-number table coerces to the rational cluster table. A further automatic-cardinality form lets the construction prove that the automatic carrier finset itself has the required cardinality lower bound; Lean then uses that finset as the lower witness. The strongest current statement builds in Lean’s canonical sorted-quad cluster witness, leaving only the canonical carrier-cardinality inequalities to be supplied on the lower side. The OEIS four-base case of those automatic cardinality inequalities is already proved from the exact five- and seven-point certificates, including the canonical all-ones relabeling used by the theorem-facing lower boundary and its local monotone lower certificates. Lean now also proves the matching upper table for this canonical all-ones base case: the concrete (Q_0, Q_1) half-line-outward route gives a direct whole-carrier ≤ 5 certificate in both orientations, and the close/non-intriguing classification of this exceptional pair is lifted through the canonical relabeling to prove that its canonical case bound is exactly 5. The generic noncoincident carrier theorem gives ≤ 7 for the other canonical base pairs. Lean also proves, from exact normalized directions, that all five non-exceptional OEIS base pairs are not close, and from strict Paulsen obtuse-distance inequalities that every distinct exact OEIS base pair is non-intriguing. These facts lift through the canonical all-ones relabeling, so every non-exceptional canonical pair has case bound exactly 7, and the canonical automatic pair table satisfies the canonical case bound for every base pair. These upper inequalities combine with the lower table to give exact equality between the canonical automatic pair table and the rational Karlsson cluster table. Lean also sums this exact automatic table to 40 and proves the corresponding base-region arithmetic $45 = \text{pairSum} + 4 + 1$. The new **TheoremOneAssembly/** folder presents the theorem-facing Lean endpoint in statement/subtheorems/proof order, and **StatementProof/EndpointTheorems.lean** re-exports the proved complete-formalization theorem endpoints under the clean statement/proof namespace. Thus future concrete blow-up geometry need not classify every extra intersection away in order to prove the lower-bound half.

Subtracting $4\binom{n}{2}$ from (24), with $n = m_1 + m_2 + m_3 + m_4$, gives

$$L(m_1, m_2, m_3, m_4) - 4\binom{n}{2} = 3 \sum_{i < j} m_i m_j - 2m_1 m_2.$$

For a fixed multiset of four cluster sizes, the product penalized by the coefficient -2 should be the smallest product. Hence the best blow-up has

$$\max L(m_1, m_2, m_3, m_4) = 4\binom{n}{2} + S(n).$$

By (1), this gives

$$a_{\mathcal{L}}(n) \geq 4 \binom{n}{2} + S(n) + n + 1. \quad (25)$$

The upper bound follows from Lemma 2, Theorem 4, and (6). Thus (25) is sharp, proving Theorem 1.

7 Numerical table and asymptotics

The continuous relaxation of (12) is minimized at

$$a = b = \frac{3n}{16}, \quad c = d = \frac{5n}{16}.$$

Consequently

$$S(n) = \frac{33}{32}n^2 + O(1),$$

and

$$a_{\mathcal{L}}(n) = 4 \binom{n}{2} + \frac{33}{32}n^2 + n + O(1) = \frac{97}{32}n^2 - n + O(1).$$

Acknowledgments

We thank David O. H. Cutler, Jonas Karlsson, Neil J. A. Sloane, and Matthias Paulsen for the lollipop problem and for the geometric ideas on which this note builds. OpenAI’s ChatGPT was used during exploratory algebra checking, finite-case audit organization, and drafting assistance.

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n	one optimal (a, b, c, d)	$S(n)$	$a_{\mathcal{L}}(n)$
0	(0,0,0,0)	0	1
1	(0,0,0,1)	0	2
2	(0,0,1,1)	3	10
3	(0,1,1,1)	9	25
4	(1,1,1,1)	16	45
5	(1,1,1,2)	25	71
6	(1,1,2,2)	37	104
7	(1,2,2,2)	50	142
8	(1,2,2,3)	65	186
9	(1,2,3,3)	83	237
10	(2,2,3,3)	103	294
11	(2,2,3,4)	124	356
12	(2,2,4,4)	148	425
13	(2,3,4,4)	174	500
14	(2,3,4,5)	201	580
15	(2,3,5,5)	231	667
16	(3,3,5,5)	264	761
17	(3,3,5,6)	297	859
18	(3,3,6,6)	333	964
19	(3,4,6,6)	372	1076
20	(4,4,6,6)	412	1193
21	(4,4,6,7)	454	1316
22	(4,4,7,7)	499	1446
23	(4,5,7,7)	545	1581
24	(4,5,7,8)	593	1722
25	(4,5,8,8)	644	1870
26	(5,5,8,8)	697	2024
27	(5,5,8,9)	751	2183
28	(5,5,9,9)	808	2349
29	(5,6,9,9)	867	2521
30	(5,6,9,10)	927	2698

Table 1: Values from the closed formula.

A Paulsen's forced intriguing pair

We recall the short linear-algebra proof that among five circles some pair is intriguing. Suppose five circles with centers $x_i \in \mathbb{R}^2$ and radii $r_i > 0$ are pairwise non-intriguing. Then every pair intersects at an obtuse angle, equivalently

$$r_i^2 + r_j^2 < |x_i - x_j|^2 < (r_i + r_j)^2. \quad (26)$$

For each circle set

$$v_i = \frac{1}{r_i}(1, r_i^2 - |x_i|^2, x_i) \in \mathbb{R}^4,$$

and endow $\mathbb{R}^4$, written as triples (α, β, x) , with the symmetric bilinear form

$$\langle (\alpha, \beta, x), (\alpha', \beta', x') \rangle = \frac{1}{2}\alpha\beta' + \frac{1}{2}\beta\alpha' + x \cdot x'.$$

A direct calculation using (26) gives

$$\langle v_i, v_i \rangle = 1, \quad -1 < \langle v_i, v_j \rangle < 0 \quad (i \neq j).$$

Five vectors in $\mathbb{R}^4$ are linearly dependent. Move the negative terms in a nontrivial relation to the other side. Both sides are nonempty: the first coordinate of every v_i is $1/r_i > 0$, so a nontrivial linear relation cannot have all coefficients with the same sign. Thus

$$\sum_{i \in P} a_i v_i = \sum_{j \in N} b_j v_j, \quad a_i, b_j > 0,$$

with P, N nonempty and disjoint. Replacing the relation by its negative if necessary, assume $|P| \leq 2$. Put $w = \sum_{i \in P} v_i$. If $|P| = 1$, then $\langle \sum_{i \in P} a_i v_i, w \rangle > 0$. If $P = \{1, 2\}$, it equals $(a_1 + a_2)(1 + \langle v_1, v_2 \rangle) > 0$. But

$$\left\langle \sum_{j \in N} b_j v_j, w \right\rangle < 0,$$

since all cross inner products are negative. This contradicts the linear relation. Therefore five pairwise non-intriguing circles cannot exist.